

高等数学(II)期末复习材料

一、定积分:

$$1. \int_{-1}^1 2f(x)dx = 18, \int_{-1}^3 f(x)dx = 4, \text{则} \int_1^3 f(x)dx = \underline{5}.$$

$$\text{解: } 2 \int_{-1}^1 f(x)dx = 18$$

$$\int_{-1}^1 f(x)dx = 9$$

$$\therefore \int_1^3 f(x)dx = \int_{-1}^3 f(x)dx - \int_{-1}^1 f(x)dx = 4 - 9 = 5.$$

二、积分上限函数求导:

$$2. \text{设 } \Phi(x) = \int_0^{x^2} \cos t dt, \text{则 } \Phi'(x) = \underline{2x \cos x^2}.$$

$$\text{解: } \Phi'(x) = \cos(x^2) \cdot (x^2)' = 2x \cos x^2.$$

$$3. \text{设 } \Phi(x) = \int_x^0 \ln(1+t^2) dt, \text{则 } \Phi'(x) = \underline{-\ln(1+x^2)}.$$

$$\text{解: } \Phi(x) = -\int_0^x \ln(1+t^2) dt$$

$$\Phi'(x) = -\ln(1+x^2)$$

三、微积分基本公式:牛顿莱布尼茨公式

$$4. \text{定积分} \int_0^2 |x-1| dx.$$

$$\text{解: } \int_0^1 (-x+1)dx + \int_1^2 (x-1)dx$$

$$= \left(-\frac{1}{2}x^2 + x\right)\Big|_0^1 + \left(\frac{1}{2}x^2 - x\right)\Big|_1^2$$

$$= -\frac{1}{2} + 1 + 0 + \frac{1}{2}$$

$$= 1$$

$$5. \text{定积分} \int_{-2}^{-1} \frac{1}{x} dx.$$

$$\text{解: } \ln|x|\Big|_{-2}^{-1} = \ln|-1| - \ln|-2| = -\ln 2$$

四、第一类换元积分法:

$$6. \text{定积分} \int_{-a}^a (2x + \sin x) dx.$$

$$\text{解: } f(-x) = -2x - \sin x = -(2x + \sin x) = -f(x)$$

$\therefore f(x)$ 为奇函数, 所以根据奇函数的积分性质其积分等于 0.

$$7. \text{定积分} \int_{-1}^1 \frac{x}{\sqrt{2+x^2}} dx.$$

$$\text{解: } f(-x) = \frac{-x}{\sqrt{2+(-x)^2}} = \frac{-x}{\sqrt{2+x^2}} = -f(x)$$

$\therefore f(x)$ 为奇函数, 所以根据奇函数的积分性质其积分等于 0.

五、第二类换元积分法:

$$8. \text{计算定积分} \int_0^{\ln 2} \sqrt{e^x - 1} dx.$$

$$\text{解: 令 } t = \sqrt{e^x - 1}, x = \ln(1+t^2), dx = \frac{2t}{1+t^2} dt, \text{当}$$

$$x=0 \text{ 时, } t=0, \text{当 } x=\ln 2 \text{ 时, } t=1.$$

$$\therefore \int_0^1 \frac{2t^2}{t^2+1} dt$$

$$= \int_0^1 \left(2 - \frac{2}{t^2+1}\right) dt$$

$$= (2t - 2\arctan t)\Big|_0^1$$

$$= 2 - \frac{\pi}{2}$$

$$9. \text{计算定积分} \int_0^4 \frac{\sqrt{x}}{1+\sqrt{x}} dx.$$

$$\text{解: 令 } t = \sqrt{x}, \text{则 } x = t^2, dx = 2t dt, \text{当 } x=0 \text{ 时, } t=0, \text{当 } x=4 \text{ 时, } t=2.$$

$$\therefore \int_0^2 \frac{2t^2}{1+t} dt$$

$$= \int_0^2 \left(2t - 2 + \frac{2}{t+1}\right) dt$$

$$= (t^2 - 2t + 2\ln(t+1))\Big|_0^2$$

$$= 4 - 4 + 2\ln 3 - 0 + 0 - 0$$

$$= 2\ln 3$$

六、分部积分法: 反对幂指三, 相乘减调个

$$10. \text{求定积分} \int_1^e 2x \ln x dx.$$

$$\text{解: } = \int_1^e x d\frac{x}{2}$$

$$\therefore u = \ln x, v' = 2x, u' = \frac{1}{x}, v = x^2$$

$$\therefore = (x^2 \ln x)\Big|_1^e - \int_1^e x dx$$

$$= e^2 - \frac{1}{2}x^2\Big|_1^e$$

$$= e^2 - \frac{e^2}{2} + \frac{1}{2}$$

$$= \frac{1}{2}e^2 + \frac{1}{2}$$

七、反常积分法:

$$11. \text{判断} \int_1^{+\infty} \frac{1}{x^p} dx \text{ 的敛散性.}$$

$$\text{解: 当 } p=1 \text{ 时, } \int_1^{+\infty} \frac{1}{x} dx = \ln|x|\Big|_1^{+\infty} = \infty - 0 = \infty (\text{发散})$$

$$\text{当 } p < 1 \text{ 时, } \int_1^{+\infty} x^{-p} dx = \frac{1}{1-p} x^{-p+1}\Big|_1^{+\infty} = \infty (\text{发散})$$

$$\text{当 } p > 1 \text{ 时, } \int_1^{+\infty} x^{-p} dx = \frac{x^{1-p}}{1-p}\Big|_1^{+\infty} = 0 - \frac{1}{1-p} = \frac{1}{p-1} (\text{收敛})$$

$$\therefore \begin{cases} p \leq 1 \text{ 发散} \\ p > 1 \text{ 收敛} \end{cases}$$

12. 判断 $\int_1^{+\infty} \frac{1}{x^4} dx$ 的敛散性.

(1) 解: $\because p = 4 > 1$ 由 p 积分判别法可知积分收敛, 且和为 $\frac{1}{p-1} = \frac{1}{4-1} = \frac{1}{3}$.

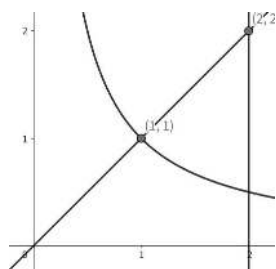
$$\begin{aligned} (2) \text{ 解: } \int_1^{+\infty} \frac{1}{x^4} dx &= \int_1^{+\infty} x^{-4} dx = \left. \frac{x^{-3}}{-3} \right|_1^{+\infty} \\ &= 0 + \frac{1}{3} = \frac{1}{3} \\ \therefore \int_1^{+\infty} \frac{1}{x^4} dx &= \frac{1}{3} \\ \therefore \text{原积分收敛且和为 } \frac{1}{3}. \end{aligned}$$

八、定积分的应用:

13. 求由曲线 $y = \frac{1}{x}$ 及直线 $y = x, x = 2$ 所围成的平面图形的面积.

$$\text{解: } \begin{cases} y = x \\ y = \frac{1}{x} \end{cases}, \text{ 解得: } \begin{cases} A_1(1, 1) \\ A_2(-1, -1) \end{cases} \text{ (舍)}$$

$$\begin{aligned} &\int_1^2 (x - \frac{1}{x}) dx \\ &= \left(\frac{x^2}{2} - \ln|x| \right) \Big|_1^2 \\ &= (2 - \ln 2) - \left(\frac{1}{2} - 0 \right) \\ &= \frac{3}{2} - \ln 2 \\ \therefore S &= \frac{3}{2} - \ln 2. \end{aligned}$$



九、微分方程基本概念:

14. 微分方程 $xy''' + y^{(4)} - xy^3 + 1 = 0$ 是 4 阶的微分方程.

解析: 判断微分方程的阶数, 只看最高阶导数, 其中 $y^{(4)}$ 为最高阶导数, 所以该微分方程是四阶微分方程.

十、一阶线性非齐次微分方程:

公式: 若: $y' + P(x)y = Q(x)$

$$\text{则有: } y = e^{-\int P(x) dx} \left[\int Q(x) e^{\int P(x) dx} dx + C \right]$$

15. 求微分方程 $y' + y = e^{-x}$ 的通解.

$$\begin{aligned} (1) \text{ 解: 公式法: } P(x) &= 1, Q(x) = e^{-x} \\ \text{代入公式有: } y &= e^{-x} \left[\int e^{-x} \cdot e^x dx + C \right] \\ &= e^{-x} \left[\int 1 dx + C \right] \\ &= e^{-x} (x + C) \end{aligned}$$

(2) 解: 常数变易法:

$$\begin{aligned} y' + y &= 0 \\ \frac{dy}{dx} &= -y \\ \int \frac{dy}{y} &= \int -dx \\ \ln y &= -x + C \\ y &= e^{-x+C} \\ y &= e^{-x} e^C \\ y &= C e^{-x} \end{aligned}$$

设 $y = C(x)e^{-x}$, 则 $y' = C'(x)e^{-x} - C(x)e^{-x}$

代入原方程有:

$$\begin{aligned} C'(x)e^{-x} - C(x)e^{-x} + C(x)e^{-x} &= e^{-x} \\ C'(x)e^{-x} &= e^{-x} \\ C'(x) &= 1 \end{aligned}$$

两边同时积分得 $C(x) = x + C$

$$\therefore y = (x + C)e^{-x}.$$

16. 求微分方程 $y' + 2xy = 4x$ 的通解.

(1) 解: 公式法: $P(x) = 2x, Q(x) = 4x$

$$\text{代入公式有: } y = e^{-\int 2x dx} \left[\int 4xe^{\int 2x dx} dx + C \right]$$

$$= e^{-x^2} \left[\int 4xe^{x^2} dx + C \right]$$

$$\text{令 } u = x^2, du = 2x dx$$

$$\therefore \int 4xe^{x^2} dx = 2 \int e^u du = 2e^u = 2e^{x^2}$$

$$\text{代入 } y = e^{-x^2} (2e^{x^2} + C) = 2 + Ce^{-x^2}$$

$$\therefore y = 2 + Ce^{-x^2}$$

(2) 解: 常数变易法:

$$\begin{aligned} y' + 2xy &= 0, \\ \frac{dy}{y} &= -2x dx \\ \ln|y| &= -x^2 + C \\ y &= e^{-x^2+C} \\ y &= e^{-x^2} e^C \\ y &= C e^{-x^2} \end{aligned}$$

设 $y = C(x)e^{-x^2}$, 则 $y' = C'(x)e^{-x^2} - 2xC(x)e^{-x^2}$

代入原方程有:

$$\begin{aligned} C'(x)e^{-x^2} - 2xC(x)e^{-x^2} + 2xC(x)e^{-x^2} &= 4x \\ C'(x)e^{-x^2} &= 4x \\ C'(x) &= 4xe^{x^2} \end{aligned}$$

两边同时积分得 $C(x) = \int 4xe^{x^2} dx$

$$\text{令 } u = x^2, du = 2x dx$$

$$\therefore C(x) = 2 \int e^u du$$

$$= 2e^u + C$$

$$= 2e^{x^2} + C$$

$$\begin{aligned} \text{代回有: } y &= (2e^{x^2} + C)e^{-x^2} = 2 + Ce^{-x^2} \\ \therefore y &= 2 + Ce^{-x^2} \end{aligned}$$

十一、二阶常系数齐次线性微分方程:

17. 在下列微分方程中, 通解为 $y = C_1 \cos x + C_2 \sin x$ 的是 (C).

- A. $y'' = y'$ B. $y'' + y' = 0$
C. $y'' + y = 0$ D. $y'' - y = 0$

$$\begin{aligned} (1) \text{ 解析: } \because y &= C_1 \cos x + C_2 \sin x \\ y' &= -C_1 \sin x + C_2 \cos x \\ y'' &= -C_1 \cos x - C_2 \sin x \\ y'' &= -y \\ y'' + y &= 0 \end{aligned}$$

(2) 解析:

$$\begin{aligned} \because y &= e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x) \\ \therefore \alpha &= 0, \beta = 1 \\ \therefore r &= \pm i \\ \therefore (r-i)(r+i) &= 0 \\ r^2 + 1 &= 0 \\ \therefore y'' + y &= 0 \end{aligned}$$

十二、多元函数微分学导数定义:

18. 函数 $z = \ln(9 - x^2 - y^2) + \sqrt{x^2 + y^2 - 1}$ 的定义域.

$$\begin{aligned} \text{解: } \begin{cases} 9 - x^2 - y^2 > 0 \\ x^2 + y^2 - 1 \geq 0 \end{cases} \text{ 解得 } \begin{cases} x^2 + y^2 < 9 \\ x^2 + y^2 \geq 1 \end{cases} \\ \therefore \{(x, y) | 1 \leq x^2 + y^2 < 9\} \end{aligned}$$

十三、多元函数的极限:

19. 极限 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 2}} (1 + xy)^{\frac{1}{x}}$.

解: 由重要极限得

$$\begin{aligned} \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 2}} \left[(1 + xy)^{\frac{1}{xy}} \right]^y \\ = \lim_{y \rightarrow 2} e^y \\ = e^2 \end{aligned}$$

20. 极限 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{1 - xy}{x^2 + y^2}$.

$$\text{解: } = \frac{1 - 0}{1} = 1$$

十四、偏导数:

21. 设函数 $z = y^x$, 则 $\left. \frac{\partial z}{\partial y} \right|_{(2,1)}$, $\left. \frac{\partial z}{\partial x} \right|_{(2,1)}$.

$$\text{解: } \frac{\partial z}{\partial x} = (y^x)'_x = y^x \ln y, \left. \frac{\partial z}{\partial x} \right|_{(2,1)} = 1^2 \ln 1 = 0$$

$$\left. \frac{\partial z}{\partial y} \right|_{(2,1)} = (y^x)'_y = xy^{x-1}, \left. \frac{\partial z}{\partial y} \right|_{(2,1)} = 2 \cdot 1^{2-1} = 2$$

十五、全微分:

22. 设函数 $z = x^3 \sin y^2$, 则 dz .

$$\begin{aligned} \text{解: } \frac{\partial z}{\partial x} &= 3x^2 \sin y^2 \\ \frac{\partial z}{\partial y} &= x^3 \cdot (\sin y^2)' = x^3 \cdot 2y \cos y^2 = 2x^3 y \cos y^2 \\ dz &= \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy \\ &= 3x^2 \sin y^2 dx + 2x^3 y \cos y^2 dy \end{aligned}$$

23. 设函数 $z = \sin xy + ye^x$, 则 dz .

$$\begin{aligned} \frac{\partial z}{\partial x} &= \cos xy \cdot y + ye^x, \frac{\partial z}{\partial y} = \cos xy \cdot x + e^x \\ dz &= \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy \\ &= (y \cos xy + ye^x) dx + (x \cos xy + e^x) dy \end{aligned}$$

十六、高阶导:

24. 设函数 $z = \ln xy + 4x^2 y^2$, 则 $\frac{\partial^2 z}{\partial x \partial y}$.

$$\begin{aligned} \text{解: } \frac{\partial z}{\partial x} &= \frac{1}{xy} \cdot y + 8xy^2 = \frac{1}{x} + 8xy^2 \\ \frac{\partial^2 z}{\partial x \partial y} &= \left(\frac{1}{x} + 8xy^2 \right)'_y = 0 + 16xy = 16xy \end{aligned}$$

25. 设函数 $z = \sin x^4 + y^4 - 4x^2 y^2$, 则 $\frac{\partial^2 z}{\partial x \partial y}$.

$$\begin{aligned} \text{解: } \frac{\partial z}{\partial x} &= \cos x^4 \cdot 4x^3 + 0 - 4y^2 \cdot 2x \\ &= 4x^3 \cos x^4 - 8xy^2 \\ \frac{\partial^2 z}{\partial x \partial y} &= (4x^3 \cos x^4 - 8xy^2)'_y \\ &= 0 - 16xy = -16xy \end{aligned}$$

十七、复合函数求导:

26. $z = f(x^3 y, xy^3)$, 其中 f 可微, 求 $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$.

$$\begin{aligned} \text{解: 设 } u &= x^3 y, v = xy^3, z = f(u, v) \\ \frac{\partial z}{\partial x} &= \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} \\ &= f'_1 \cdot 3x^2 y + f'_2 \cdot y^3 \\ &= 3x^2 y f'_1 + y^3 f'_2 \\ \frac{\partial z}{\partial y} &= \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} \\ &= f'_1 \cdot x^3 + f'_2 \cdot 3xy^2 \\ &= x^3 f'_1 + 3xy^2 f'_2 \end{aligned}$$

27. $z = u^2 v, u = e^t, v = \sin t$, 求 $\frac{dz}{dt}$.

$$\begin{aligned} \text{解: } \frac{dz}{dt} &= \frac{\partial z}{\partial u} \frac{du}{dt} + \frac{\partial z}{\partial v} \frac{dv}{dt} \\ &= 2uv \cdot e^t + u^2 \cdot \cos t \\ &= 2e^t \sin t \cdot e^t + e^{2t} \cdot \cos t \end{aligned}$$

$$= e^{2t}(2\sin t + \cos t)$$

十八、隐函数求导:

28. 已知 $e^z - xyz = 1$, 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

解: 设 $F(x, y, z) = e^z - xyz - 1$

$$F'_x = -yz, F'_y = -xz, F'_z = e^z - xy$$

$$\frac{\partial z}{\partial x} = -\frac{F'_x}{F'_z} = \frac{yz}{e^z - xy}$$

$$\frac{\partial z}{\partial y} = -\frac{F'_y}{F'_z} = \frac{xz}{e^z - xy}$$

29. 已知 $\frac{x}{z} = \ln \frac{z}{y}$, 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

解: 设 $F(x, y, z) = \frac{x}{z} - \ln \frac{z}{y}$

$$F'_x = \frac{1}{z}, F'_y = 0 - \frac{y}{z} \cdot z \cdot \left(-\frac{1}{y^2}\right) = \frac{1}{y}$$

$$F'_z = x \cdot \left(-\frac{1}{z^2}\right) - \frac{y}{z} \cdot \frac{1}{y} = -\frac{x+z}{z^2}$$

$$\frac{\partial z}{\partial x} = -\frac{F'_x}{F'_z} = -\frac{\frac{1}{z}}{-\frac{x+z}{z^2}} = \frac{1}{z} \cdot \frac{z^2}{x+z} = \frac{z}{x+z}$$

$$\frac{\partial z}{\partial y} = -\frac{F'_y}{F'_z} = -\frac{\frac{1}{y}}{-\frac{x+z}{z^2}} = \frac{z^2}{y(x+z)}$$

十九、多元函数最值:

30. 要制造一个容积为 V 的长方体箱子, 问如何设计才能使所用材料最少?

解: 设长方体箱子的长、宽、高分别为 x, y, z .

目标函数: $S = 2(xy + xz + yz)$

约束条件: $xyz = V$

构造 $L = 2(xy + xz + yz) + \lambda(xyz - V)$

$$\text{令} \begin{cases} \frac{\partial L}{\partial x} = 2y + 2z + \lambda yz = 0 \text{ ①} \\ \frac{\partial L}{\partial y} = 2x + 2z + \lambda xz = 0 \text{ ②} \\ \frac{\partial L}{\partial z} = 2x + 2y + \lambda xy = 0 \text{ ③} \\ \frac{\partial L}{\partial \lambda} = xyz - V = 0 \text{ ④} \end{cases}$$

$$\begin{aligned} \text{①} - \text{②} &= 2y + 2z + \lambda yz - (2x + 2z + \lambda xz) = 0 \\ &2(y - x) + \lambda z(y - x) = 0 \\ &(y - x)(2 + \lambda z) = 0 \end{aligned}$$

$$\text{同理 } \text{②} - \text{③} = (x - z)(2 + \lambda y) = 0$$

$$\text{①} - \text{③} = (y - z)(2 + \lambda x) = 0$$

$$\begin{cases} (y - x)(2 + \lambda z) = 0 \\ (x - z)(2 + \lambda y) = 0, \therefore x = y = z \text{ 为唯一驻点} \\ (y - z)(2 + \lambda x) = 0 \end{cases}$$

代入 ④ 有 $x = \sqrt[3]{V}, y = \sqrt[3]{V}, z = \sqrt[3]{V}$,

$\therefore$ 当长、宽、高均为 $\sqrt[3]{V}$ 时所用材料最少.

二十、二重积分的性质:

31. 若在 D 上, $f(x, y) = 2$, 记 $\mu(D)$ 为 D 的面积, 则 $\iint_D f(x, y) d\sigma$.

解: 将 $f = 2$ 代入 $\iint_D f(x, y) d\sigma$.

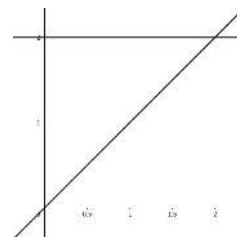
$$\text{得 } \iint_D 2 d\sigma = 2 \iint_D d\sigma$$

$$\text{又 } \because \iint_D d\sigma = \mu(D)$$

$$\therefore \iint_D f(x, y) d\sigma = 2\mu(D)$$

32. 设 D 域为 $0 \leq x \leq y, 0 \leq y \leq 2$, 则 $\iint_D d\sigma$.

$$\begin{aligned} \text{解: } \iint_D d\sigma &= \int_0^2 \int_0^y 1 dx dy \\ &= \int_0^2 (x|_0^y) dy \\ &= \int_0^2 y dy = \frac{y^2}{2} \Big|_0^2 \\ &= 2 \end{aligned}$$



二十一、交换积分次序:

33. 交换二次积分的积分次序

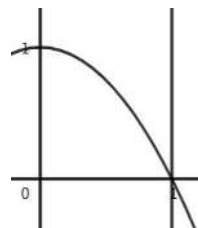
$$\int_0^1 dy \int_0^{\sqrt{1-y}} f(x, y) dx.$$

$$\text{解: } D = \begin{cases} 0 \leq y \leq 1 \text{ ①} \\ 0 \leq x \leq \sqrt{1-y} \text{ ②} \end{cases}$$

将 ② 变形得 $y \leq 1 - x^2$

$$\text{由图知 } D = \begin{cases} 0 \leq x \leq 1 \\ 0 \leq y \leq 1 - x^2 \end{cases}$$

交换积分次序得 $\int_0^1 dx \int_0^{1-x^2} f(x, y) dy$



二十二、二重积分计算:

34. 计算二重积分 $\iint_D e^{y^2} d\sigma$, 其中 D 由 $y = x, y = 1$ 及 y 轴围成的闭区域.

解: 画图可知 $D = \begin{cases} 0 \leq y \leq 1 \\ 0 \leq x \leq y \end{cases}$

$$\begin{aligned} &\int_0^1 dy \int_0^y 2e^{y^2} dx \\ &= \int_0^1 dy (2e^{y^2} \cdot x|_0^y) \\ &= \int_0^1 dy \cdot 2e^{y^2} \cdot y \\ &= \int_0^1 2e^{y^2} \cdot y dy \\ &= \int_0^1 e^{y^2} dy^2 \text{ (第一类换元积分)} \\ &= e^{y^2} \Big|_0^1 \\ &= e - 1 \end{aligned}$$

